Day 15 Notes IAM (Identity and Access Management)

Introduction to IAM

IAM stands for Identity and Access Management.

IAM is an AWS service used to:

- Create users
- Manage permissions
- Control access to AWS resources
- Improve security in AWS environment

IAM helps organizations securely manage who can access AWS services and what actions they can perform.

Why IAM is Important

In a company:

- Different employees require different access.
- Developers may need EC2 access.
- Database teams may need RDS access.
- Finance team may need Billing access.

IAM helps provide:

- Secure access
- Limited permissions
- Centralized management

Authentication and Authorization

Authentication

Authentication means:

Who are you?

Example:

- Username
- Password
- MFA

Authorization

Authorization means:

What can you access?

Example:

- EC2 access
- S3 access
- Read-only access

IAM Components

IAM mainly contains:

- Users
- Groups
- Policies
- Roles

IAM Users

IAM User is an individual AWS account identity created for a person.

Each user can have:

- Username
- Password
- Access keys

- Permissions

Example

User	Access
Student1	EC2 Access
Admin	Full Access
Developer	S3 Access

IAM Groups

IAM Group is a collection of IAM users.

Groups simplify permission management.

Instead of assigning permissions to every user individually:

- Create a group
- Assign permissions to group
- Add users into group

Example

Group Name	Permission
DevOps-Team	EC2 Full Access
S3-Team	S3 Access
Admin-Team	Administrator Access

Advantages of Groups

- Easy management
- Centralized permissions

- Reduces manual work
- Better administration

IAM Policies

Policies are permission documents written in JSON format.

Policies define:

- Which AWS services can be accessed
- What operations are allowed
- Which resources can be used

Example Policy

```
{
  "Effect": "Allow",
  "Action": "ec2:*",
  "Resource": "*"
}
```

Policy Keywords

Keyword	Meaning
Effect	Allow or Deny
Action	AWS operation
Resource	AWS resource

IAM Roles

IAM Role is one of the most important concepts in AWS.

Definition

IAM Role provides temporary permissions to AWS services or users.

Roles are mainly used by:

- EC2
- Lambda
- ECS
- AWS services

Why Roles are Required

Suppose an EC2 instance needs access to S3 bucket.

Without role:

- Access Key and Secret Key must be stored inside EC2 server.
- This creates security risks.

AWS recommends using IAM Roles instead of storing credentials manually.

Real-Time Example

EC2 Accessing S3 Bucket

Scenario

An EC2 server needs to:

- Read files from S3 bucket
- Upload logs to S3

Instead of storing passwords inside server:

- Create IAM Role
- Attach S3 permissions
- Attach role to EC2

AWS automatically provides temporary credentials to EC2.

Role Architecture

EC2 Instance
↓
IAM Role
↓
S3 Bucket Access

How IAM Roles Work

Step 1

Create IAM Role.

Step 2

Attach required permission policy.

Example:

AmazonS3ReadOnlyAccess

Step 3

Attach role to EC2 instance.

Step 4

AWS automatically provides temporary credentials to EC2.

Benefits of IAM Roles

Benefit	Description
More Secure	No passwords stored
Temporary Credentials	Auto-managed by AWS
Easy Management	Centralized access

Benefit	Description
Best Practice	AWS recommended approach

IAM User vs IAM Role

IAM User	IAM Role
Permanent identity	Temporary permissions
Username and password	Temporary credentials
Mainly for humans	Mainly for AWS services
Long-term access	Short-term access

Session Manager Role Example

AWS Session Manager allows connecting to EC2 without:

- SSH
- Port 22
- Key pair

Required Policy:

AmazonSSMManagedInstanceCore

Benefits:

- Secure access
- Browser-based terminal
- No SSH management

Best Practices in IAM

- Avoid using Root account
- Enable MFA

- Follow Least Privilege Principle
- Use Groups for permission management
- Use Roles for AWS services
- Rotate credentials regularly

Least Privilege Principle

Provide only required permissions.

Example:

- If user needs only S3 access,
  do not provide Administrator access.

Real-Time Industry Usage

Service	IAM Role Usage
EC2	Access S3
Lambda	Access DynamoDB
ECS	Pull Docker images
CloudWatch	Send monitoring logs

Important Commands

Verify Role in EC2

aws sts get-caller-identity

List S3 Buckets

aws s3 ls

Class Summary

IAM manages authentication and authorization in AWS.
IAM Users are permanent identities.
IAM Groups simplify permission management.
IAM Policies define permissions.
IAM Roles provide temporary AWS permissions securely.

Interview Questions

1. What is IAM?
2. Difference between Authentication and Authorization?
3. What is IAM User?
4. What is IAM Group?
5. What is IAM Policy?
6. What is IAM Role?
7. Difference between IAM User and IAM Role?
8. Why are IAM Roles more secure than Access Keys?
9. What is Least Privilege Principle?
10. What is the use of Session Manager?
11. Which policy is required for Session Manager?
12. Can EC2 access S3 without access keys?
13. Why should Root account usage be avoided?
14. What are AWS managed policies?
15. Explain real-time IAM Role usage in EC2.